McGill | Department of Epidemiology, Biostatistics and Occupational Health

BIOS 643 - Health Data Analysis: Thry to Prac - Analysis & Rptg

Assessment Week 2

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Data Cleaning Plan

Cohort Dataset

Issues found from data overview and how corrected

- The variables entry date and lost to follow-up date had some values that could not be recognized as dates.
  - The parse_date_time() function of the lubridate package was used to identify and validate all date formats.
  - The original entry date and lost to follow-up variables were then dropped after creating new variables using the parse_date_time() function of the lubridate package.
- Column names were inconsistent with a mix of lower and upper cases with no structured format
  - The dplyr package was used to rename specific columns
  - The janitor package was used to standardize variable names
  - Columns were finally logically arranged with the select() function
- Duplicated row check
  - Three (3) duplicated rows found; specifically id: 38, 1782 and 1744
  - These were then removed to have distinct rows
- The skim() function from the skimr package showed no missing values
- The variables id, age, sex, bmi are numeric from the skim() function of the skimr package
  - sex was therefore recoded as factor
- Smoking status was not accurately labelled (mix of upper and lower cases)
  - Mutated to have unique labels and recoded as a factor

- Body mass index (bmi) was categorised into underweight, normal, overweight and obese and then orderd as a factor variable
- Socio-economic status (ses_status) was also ordered as as factor variable
- Application of the minimum age at entry excluded 8 participants from the cohort.
- The cleaned cohort dataset was then exported for sharing and future analysis.

```
##
##   Never  Former Current
##     561     231     105
```

```
##
##      Normal       Obese  Overweight Underweight
##         303         212         329          53
```

```
##
##    Low Medium   High
##    259    445    193
```

```
## [1] TRUE
```

```
## [1] 3
```

```
##       id entry_date age    sex  bmi    bmi_cat ses_status smoking_status
## 895   38 1990-02-15  52 Female 18.7     Normal        Low          Never
## 896 1782 2006-04-08  69 Female 19.5     Normal     Medium          Never
## 897 1744 2005-01-16  60 Female 27.9 Overweight     Medium         Former
##      lost_date
## 895 1999-08-13
## 896 2015-10-04
## 897 2007-12-27
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   33.00   55.00   62.00   61.74   68.00   85.00
```

icd dataset

Issues found from data overview and how corrected

- The variable "description" was renamed to icd_description
- Duplicated row check showed no duplicated rows
- skim() function from the skimr package showed no missing values
- Multiple diagnostic dates (icd_date) for the same person (id) for the same disease(icd_code):
  - slice_min() function of the dplyr package was used to only retain the first diagnosis.
- COPD outcome indicator was created using the mutate() function and recoded as a factor, where 0 means copd absent (No) and 1 means copd present (Yes)
- A copd comorbidy indicator was also defined using the summary() function.
  - The comorbidity variable was recoded as a factor where 0 means absence of comorbidity (No) and 1 means the presence of comorbidity (Yes)
  - This mini dataset was then merged to the original to have a composite icd dataset.
- The cleaned icd dataset was then exported for sharing and future analysis

```
## [1] FALSE
```

```
##
##   No  Yes
## 3337  186
```

```
##
##    0    1
## 2968  112
```

```
##
##   No  Yes
## 2968  112
```

residential dataset

Issues found from data overview and how corrected

- skim() function from the skimr package showed no missing values
- Duplicated row check showed no duplicated rows
- All dates were ensured they were date objects from the as.Date() function of the lubridate package
- First residential address dates were grouped by particpant id and then summerised to calculate the number of years between first residential address date and cohort entry date (lag_years)
- The mini dataset was then merged to the original residential dataset, selecting only id and lag_years variables from the mini dataset
- Application of a minimum of 10 year lag_date (i.e period between participant's first residential address and cohort entry date) did not exclude any participant, as the minimum number of years found was 10.10 years from summary statistics.
- Variables were finally arranged in a logical format
- The cleaned residential dataset was was exported for sharing and future analysis.

```
## [1] FALSE
```

```
## [1] "id"                 "address_start_date" "entry_date"
## [4] "address_end_date"   "state"              "county"
## [7] "lag_years"
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   10.10   13.50   17.01   18.38   23.15   30.20
```

Combined dataset

- The as.Date() function of the lubridate package was used to ensure all dates are date objects
- Duplicated row check showed no duplicated rows
- The cleaned cohort dataset was first merged with the cleaned residential dataset using id and entry dates as key
- The resulting dataset was then merged with the cleaned icd dataset using id as key
- The variables were then rearranged using the select() function in a logical order
- Some variable names were cleaned and standardized
- skim() function from the skimr package showed that there are a total of 90 missing values from the following variables: addr_start_date, addr_end_date and lag_years.
- The final cleaned and merged dataset was exported for sharing and further analysis.

```
## [1] FALSE
```

```
## [1] FALSE
```

```
##  [1] "id"              "entry_date"      "lag_years"       "age"
##  [5] "sex"             "bmi"             "bmi_cat"         "ses_status"
##  [9] "smoking_status"  "lost_date"       "county"          "state"
## [13] "addr_start_date" "addr_end_date"   "ehr_entry_date"  "icd_code"
## [17] "icd_description" "icd_date"        "copd_outcome"    "copd_comorb"
```

Quality control checks

1) Analysability

- First role has only variables
- There is participant id which serves as a unique identifier of participants

2) Interpretability

- All variable names are unique and clearly readable
- The longest variable has 16 characters which by far less than the maximum required number of character (32)

3) Completeness

- No missingness
- sample size = 1764, which is consistent from the output from the dim() function
- Distinct rows (i.e. no duplicated values)

4) Validity

- Date object have been ascertained
- Numeric variables are numeric as expected
- Categorical variables and factors are as expected

5) Accuracy

- Overview of the data for now shows accuracy

6) Consistency

- Value labeles like dates are uniformly applied, as well as categorical variables.'

7) De-identification

- As it stands now, it is not possible to identify any specific individuals from the dataset.

Data dictionary

Variable name	Data type	Values	Notes
id	Numeric	1 to inf	Unique identifier
entry_date	Date	YY-MM-DD	Date of cohort entry
age	Numeric	18 and above	Age of the respondant
sex	Factor	0=Female; 1=Male	Biological sex
bmi	Numeric	17 and above	Body mass index
bmi_cat	Factor	Underweight	bmi < 18.5
.	.	Normal	bmi 18.5 -24.9
.	.	Overweight	bmi 25-29.9
.	.	Obese	bmi > 30.0
ses_status	Factor	High/Medium/Low	Socio-economic status
smoking_status	Factor	Never/Former/Current	Smoking status
lost_date	Date	YY-MM-DD	Date lost to follow-up
county	Character	US counties	Counties in the US
state	Character	US states	States in the US
addr_start_date	Date	YY-MM-DD	Start date of residence
addr_end_date	Date	YY-MM-DD	End date of residence
ehr_entry_date	Date	YY-MM-DD	EHR entry date record
icd_code	Character	Alphanumeric	Disease diagnostic codes
icd_description	Character	-	Disease condition
icd_date	Date	YY-MM-DD	Disease diagnosis date
copd_outcome	Factor	0=No; 1=Yes	COPD outcome diagnosis
copd_comorb	Factor	0=No; 1=Yes	COPD comorbidity diagnosis
lag_years	Numeric	integer	First residence-entry date